DRIVEABILITY—BI-FUEL—PARAMETER IDENTIFICATION DATA (PID) INPUT AND OUTPUT INFORMATION—VEHICLES EQUIPPED WITH 5.4L OR 6.8L BI-FUEL ENGINE ONLY	Article No. 01-10-1

FORD: 1998-2001 E-250, E-350, F-150, F-250 LD
1999-2001 SUPER DUTY F SERIES

ISSUE

This article can be used as an overview of bi-fuel LPG vehicle Parameter Identification Data (PID), inputs and outputs as a diagnostic aid for the listed bi-fuel vehicles.

ACTION

Refer to the Parameter Identification Data (PID) Input And Output Information Chart as an aid for diagnostics and the Switch-Over Code Chart for additional information.

SERVICE INFORMATION

Refer to the PID Information table and PID Input/Output diagram (Figure 1) as an aid for diagnostics.

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PID INPUT AND OUTPUT INFORMATION			
PID Sensor/ Input	**PID State - KOEO**	**PID State - Hot Idle in Alt Fuel**	**PID - Units**
ADPTCTL	0.0	1.00 +/- 0.30 equivalent to long term fuel trim	
ALTRFUN	Counts up for total run time on alt fuel	Counts up for total run time on alt fuel	hours
BAROHRE	13.5 to 15.0 NOTE: This will vary according to local altitude and weather conditions.	13.5 to 15.0	psi
CL-COR	0.0	0.70 to 1.30	#
CLSD-LP	NO	YES	
DELTMAP	0.0	- 7.5 to +7.5 (+ reading indicates that the throttle is opening and MAP is increasing, - reading indicates the reverse)	psi
ECT	-38 to 254	175 to 254 (from PCM)	degrees F
EGRFLW	0.00	0.0 NOTE: Calculated EGR Flow. Will increase with increased EGR.	lbs/min
FRPREAB	12 to 45 NOTE: the minimum pressure to start on the alternative fuel is 26 psia for LPG. The minimum reading for a functional sensor with the inlet hose disconnected would be the same as BAROHRE.	25 to 45 NOTE: Once the engine has started in Alt Fuel, it will continue to run until readings drops below 26 psia for LPG. Below this, the engine will autoswitch to gasoline.	psia
FRTMPAB	-79 to 260	-79 to 260 NOTE: Will vary dependent on ambient temperature, engine temperature, fuel temperature and fuel demand.	degrees F
FTP	0	0 (FTP is for CNG only)	psi
FTTMP	CNG only (LPG will show -42)		degrees F
FUEL	GAS/ALT		
FUEL-MD	01, 03	01, gasoline, 02, in transition from Alt to Gasoline, 03, Alt Fuel	#
GASFRUN	0 to 65535	Counts up for total run time on gasoline	hours
HO2S	0.0	1.0/.9	
IGN-SW	ACCY, RUN	RUN	
IAT	-38 to 254	Varies according to ambient air temperature and underhood temperatures (from PCM).	degrees F
MAF	0.0	4.8 to 8.2	lbs/min
MAP 3 TO 7	14.5 NOTE: KOEO, MAP reading should equal BAROHRE reading.	NOTE: Will vary according to engine speed and load.	psia
MIX-TMP	Ambient	-40 to 215	degrees F
O2S11	0.0	0.1 to 0.9	volts
O2S21	0.0	0.1 to 0.9	volts
OPR-MD	STALL, RUN	RUN	

PID Sensor/ Input	PID State - KOEO	PID State - Hot Idle in Alt Fuel	PID - Units
RPM	0	650 to 950 at idle	
RUNTIME	0	0 to 65535 (counts up from last start)	seconds
SEL-SW	GAS/ALT	GAS/ALT	
VBAT	12 (batt volt)	11.5 to 15.0 NOTE: This is the voltage as seen by the AFCM.	volts
VCLEXP	NO/YES	YES	
VCLPRS	NO/YES	YES	
VOL-EFF	0.0	0.0 to 1.50	%
CANVTDC	%	%	
CAUSE	00, 02, 03, 04, 05, 06, 07, 08	00, 02, 03, 04, 05, 06, 07, 08 NOTE: Indicated switch-over cause. 00 = no switch-over requested. See following switch-over chart for complete list.	
CCNT	0 to 29	Number of DTCs stored	
FPLS1	0.0	0.0 to 0.02	milliseconds
FPLS2	0.0	0.0 to 0.02	milliseconds
GAS FPR	OFF/ON	Off when running on Alt Fuel, On when running on gasoline	NOTE: May not respond.
MFDES	0.00	Desired Mass Flow of Fuel. This will vary according to engine speed and load.	lbs/min
SOFFSOL	CLOSED	CLOSED	
SPRKADD	0.0	# of degrees requested beyond normal PCM timing advance.	degrees
TDCPASS	ON	ON	
VLV #1	CLOSED	OPEN/CLOSED	Injector Status
VLV #2	CLOSED	CLOSED	Injector Status
VLV #3	CLOSED	CLOSED	Injector Status
VLV #4	CLOSED	CLOSED	Injector Status
VLV #5	CLOSED	CLOSED	Injector Status
VLV #6	CLOSED	CLOSED	Injector Status

SWITCH-OVER CODE CHART		
Code #	**Switch-Over Cause**	**Description**
00	No switch-over requested	On key on, Alt fuel has been selected and there is sufficient fuel to start and run on Alt Fuel.
02	Low fuel absolute pressure	While running on Alt Fuel, the pressure dropped below the minimum to run on Alt Fuel, resulting in a switch to gasoline.
03	Requested by VCL	VCL was unable to communicate with EEC to request Alt Fuel, therefore, switch to gas.
04	Vaporizer lockout	MST below the minimum requirement to start on LPG.
05	Liquid LPG	FAP and FRT are within limits that could allow liquid LPG to reach fuel injectors.
06	Fuel absolute pressure time out	FAP did not reach the minimum level within time limit on key up. Most likely cause, out of fuel before starting.
07	Stall after auto-switch	After auto-switching to gasoline, the engine has stalled. Therefore, if the engine is cranked without switching off the key, start on gasoline.
08	No VCL communication	The Compuvalve could not communicate with the VCL or VCLD.

OTHER APPLICABLE ARTICLES: NONE
WARRANTY STATUS: INFORMATION ONLY
OASIS CODES: 490000, 499000, 603300, 608000, 611000, 690000, 698298

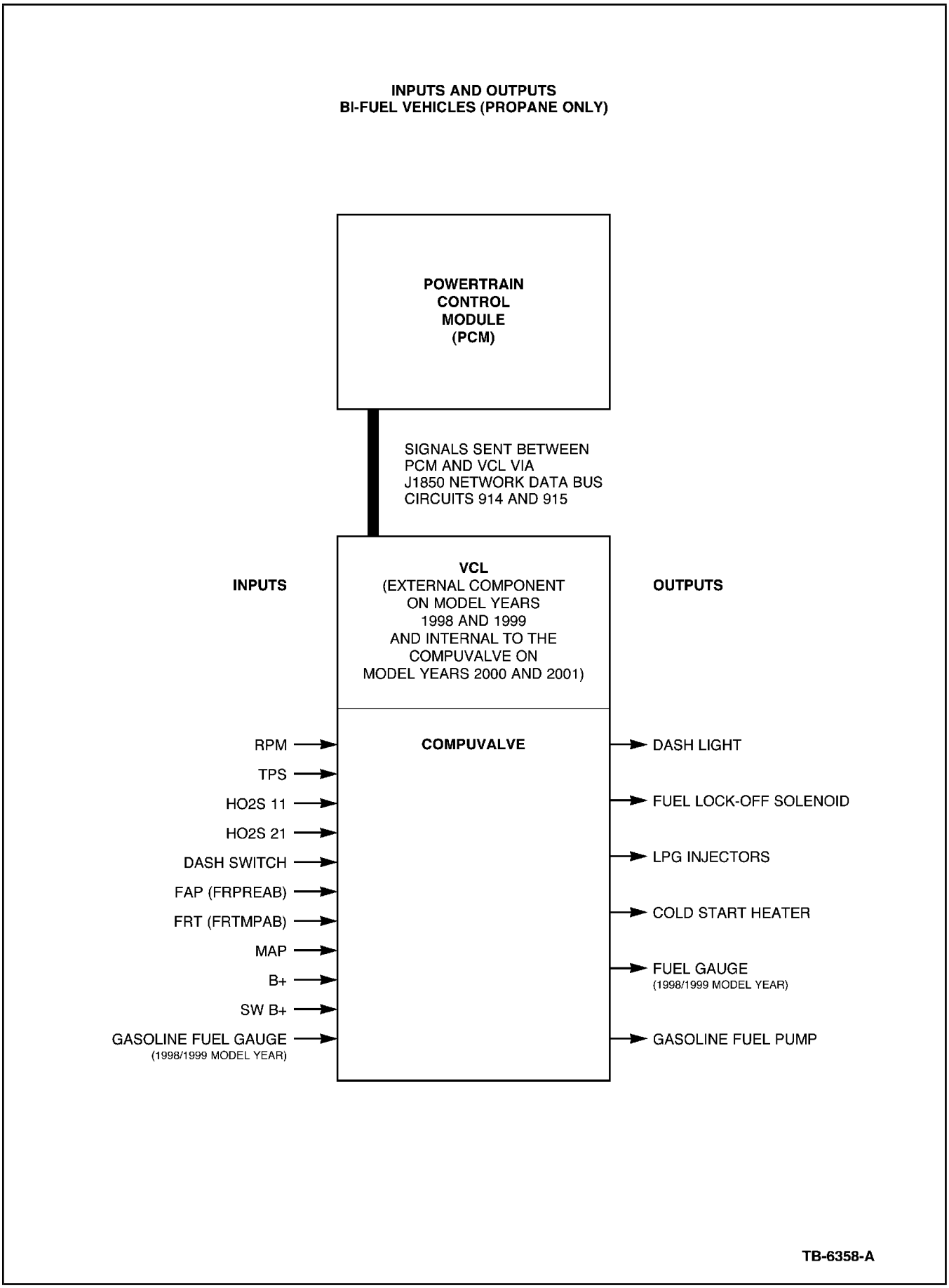

Figure 1 - Article 01-10-1